AN OBJECT OF THE DERIVED CATEGORY HAS A LEFT DUAL PRECISELY WHEN IT IS PERFECT, AND THE DUAL IS THE DUAL OF A PERFECT COMPLEX

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot; nothing was added, removed or redefined and no package was added.

Editorial note. The source URL is the source-level Stacks Project URL, as required when more than one card is produced from one source; the per-claim tag is carried by claim_key.

Editorial note. The author's typo 'Choose a K-flat complexes N^bullet' is reproduced unchanged. The references inside the counted proof span are recorded as non-load-bearing navigation pointers: categories-definition-dual and derived-lemma-subcategory-left-resolution resolve to the project's cross-chapter tag list, while lemma-dual-perfect-complex, lemma-left-dual-complex, lemma-K-flat-resolution, lemma-complex-bounded-above-free-colim-bounded-finite-free and lemma-summands-perfect are same-chapter labels resolved to their \label lines in the pinned source.

Allowed standard prerequisites (editorial).

- Left duals in a monoidal category and their unit and counit. the definition invoked at statement lines 38585-38586 and at proof line 38602. Supplies the maps eta and epsilon whose identities drive both factoring arguments.
- K-flat complexes with flat terms and the identification of the derived tensor product with a total complex. used at proof lines 38605-38621. Makes Tot(M^bullet tensor_R N^bullet) a representative of M tensor^L N and lets a finite description of eta(1) be extracted.
- Bounded above complexes of free modules and bounded complexes of finite free modules. used at proof lines 38641-38650. Provides the re-representation of M that the second factoring argument cuts down to a perfect subcomplex.
- Direct summands of perfect complexes are perfect. invoked at proof lines 38652-38653. Upgrades the factorisation of the identity through a perfect K to perfectness of M.

Source reference. Original lemma 0FFP: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 19294–19294. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 05T7: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7471–7471. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 14380: sources/stacks-more-algebra/pinned-original-source.tex, source lines 14380–14380. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 38553: sources/stacks-more-algebra/pinned-original-source.tex, source lines 38553–38553. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 19176: sources/stacks-more-algebra/pinned-original-source.tex, source lines 19176–19176. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 18462: sources/stacks-more-algebra/pinned-original-source.tex, source lines 18462–18462. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 18963: sources/stacks-more-algebra/pinned-original-source.tex, source lines 18963–18963. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let R be a ring. Let M be an object of $D(R)$. The following are equivalent*

1. *M has a left dual in $D(R)$ as in Categories, Definition 0FFP,*
2. *M is a perfect object of $D(R)$.*

Moreover, in this case the left dual of M is the object $M^\vee$ of Lemma 19176.

Proof. If M is perfect, then we can represent M by a bounded complex $M^\bullet$ of finite projective R-modules. In this case $M^\bullet$ has a left dual in the category of complexes by Lemma 18462 which is a fortiori a left dual in $D(R)$.

Assume (1). Say N, $\eta : R \to M \otimes^{\mathbf{L}}_R N$, and $\epsilon : M \otimes^{\mathbf{L}}_R N \to R$ is a left dual as in Categories, Definition 0FFP. Choose a complex $M^\bullet$ representing M. Choose a K-flat complexes $N^\bullet$ with flat terms representing N, see Lemma 14380. Then η is given by a map of complexes

$$\eta : R \longrightarrow \mathrm{Tot}(M^\bullet \otimes_R N^\bullet)$$

We can write the image of 1 as a finite sum

$$\eta(1) = \sum\nolimits_n \sum\nolimits_i m_{n,i} \otimes n_{-n,i}$$

with $m_{n,i} \in M^n$ and $n_{-n,i} \in N^{-n}$. Let $K^\bullet \subset M^\bullet$ be the subcomplex generated by all the elements $m_{n,i}$ and $\mathrm{d}(m_{n,i})$. By our choice of $N^\bullet$ we find that $\mathrm{Tot}(K^\bullet \otimes_R N^\bullet) \subset \mathrm{Tot}(M^\bullet \otimes_R N^\bullet)$ and $\eta(1)$ is in the subcomplex by our choice above. Denote K the object of $D(R)$ represented by $K^\bullet$. Then we see that η factors over a map

$\tilde{\eta} : R \longrightarrow K \otimes^{\mathbf{L}}_R N$. Since $(1 \otimes \epsilon) \circ (\eta \otimes 1) = \mathrm{id}_M$ we conclude that the identity on M factors through K by the commutative diagram

$$\begin{array}{ccccc} M & \xrightarrow{\eta\otimes 1} & M \otimes^{\mathbf{L}}_R N \otimes^{\mathbf{L}}_R M & \xrightarrow{1\otimes\epsilon} & M \\ & \searrow^{\tilde{\eta}\otimes 1} & \uparrow & & \uparrow \\ & & K \otimes^{\mathbf{L}}_R N \otimes^{\mathbf{L}}_R M & \xrightarrow{1\otimes\epsilon} & K \end{array}$$

Since K is bounded above it follows that $M \in D^-(R)$. Thus we can represent M by a bounded above complex $M^\bullet$ of free R-modules, see for example Derived Categories, Lemma 05T7. Write $\eta(1) = \sum_n \sum_i m_{n,i} \otimes n_{-n,i}$ as before. By Lemma 38553 we can find a subcomplex $K^\bullet \subset M^\bullet$ containing all the elements $m_{n,i}$ which is bounded and consists of finite free R-modules. As above we find that the identity on M factors through K. Since K is perfect we conclude M is perfect too, see Lemma 18963. $\square$